

Analysis of the Impact of Traffic Closures on Transport Using Simulations

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MOTIVATION

Traffic closures redistribute flows network-wide, causing congestion far from the closed area. This open-source pipeline enables What-if analysis of planned closures in Brno using only open data (OSM, Czech census SLD8 2021, CSD 2025 traffic counts).

Figure 1: Pipeline Overview

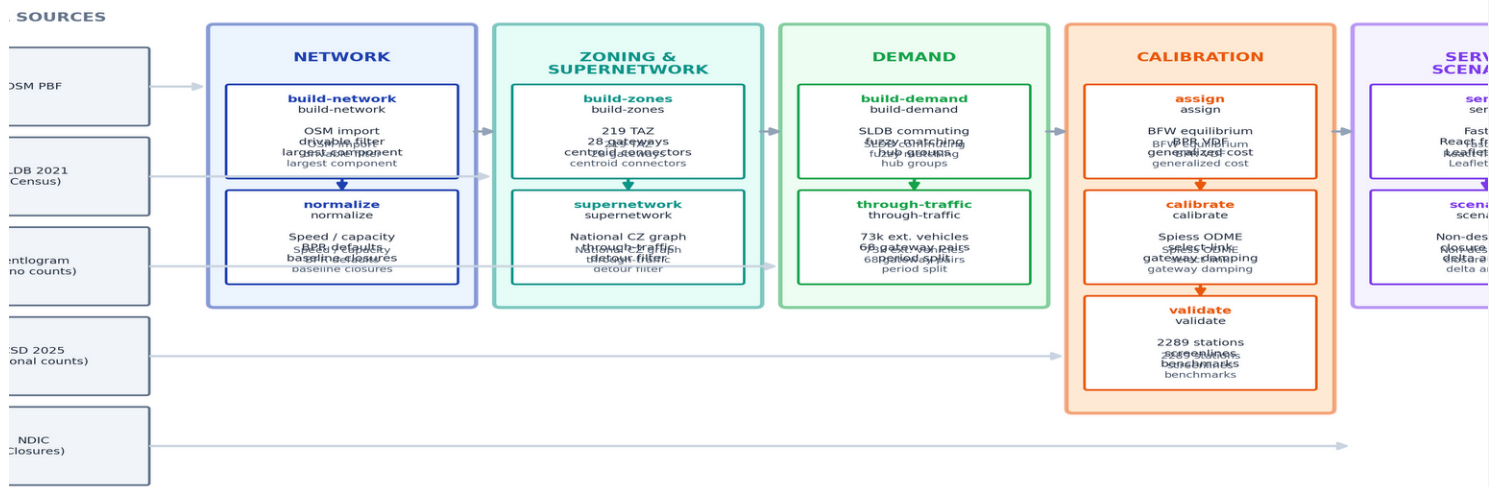
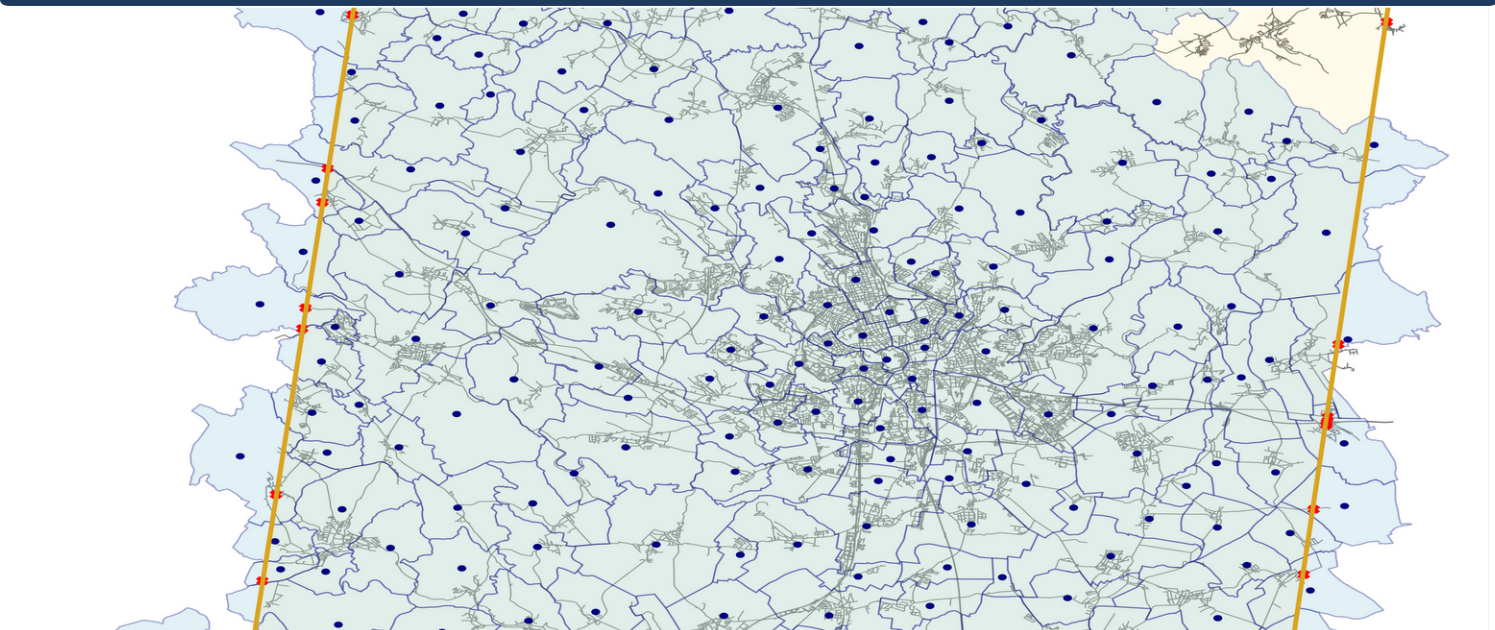


Figure 1: Pipeline Overview — 16-step deterministic pipeline from OSM import to interactive scenario engine

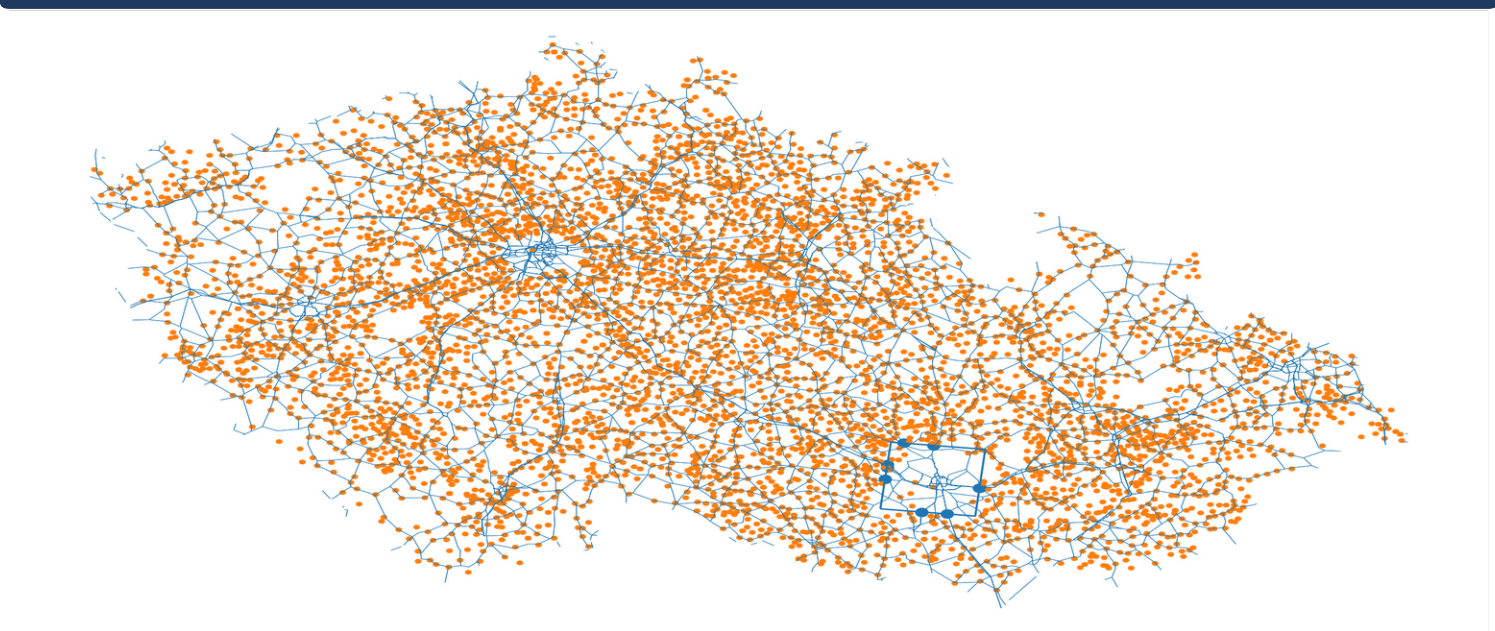
16-step deterministic pipeline: 5 color-coded phases

Figure 2: Model Area — 219 TAZ + 28 Gateways



191 Internal + 28 gateway zones, Brno + 2km

Figure 3: National Supernetwork



6117 places + 9 gateways, detour ratio ≤ 1.25, 73k veh/day

Figure 4: Scenario Methodology

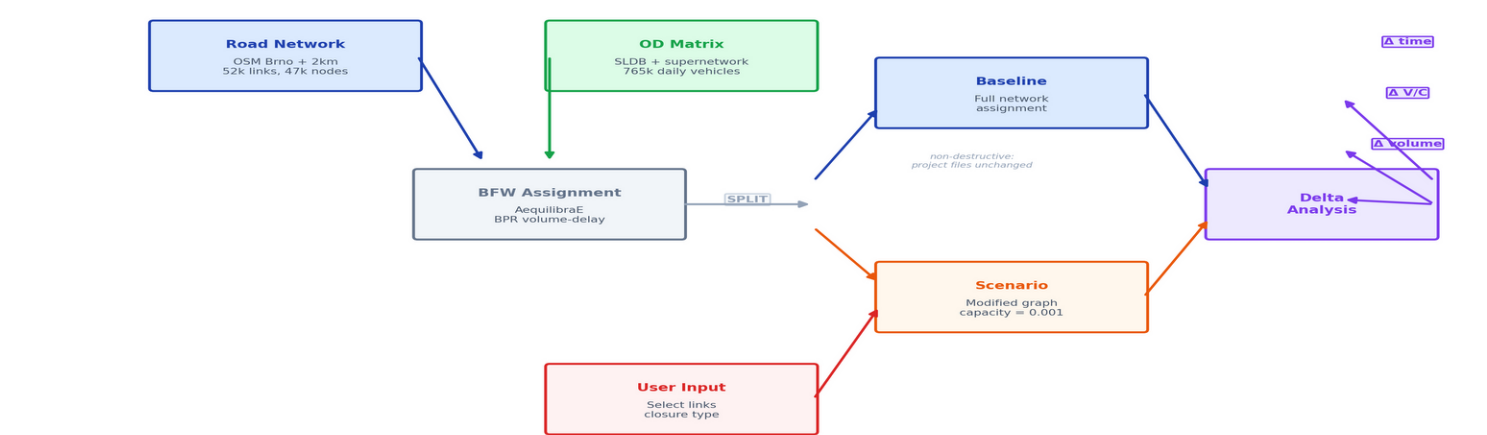
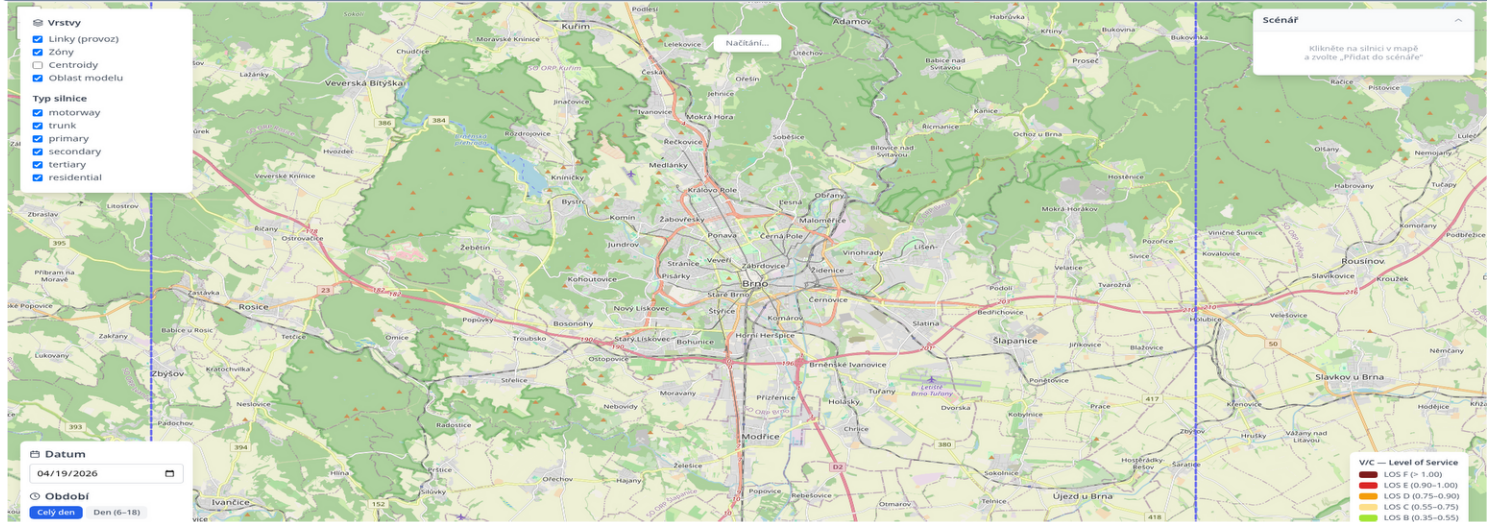


Figure 4: Scenario Methodology — Shared inputs, diverging at in-memory graph modification, per-link delta output

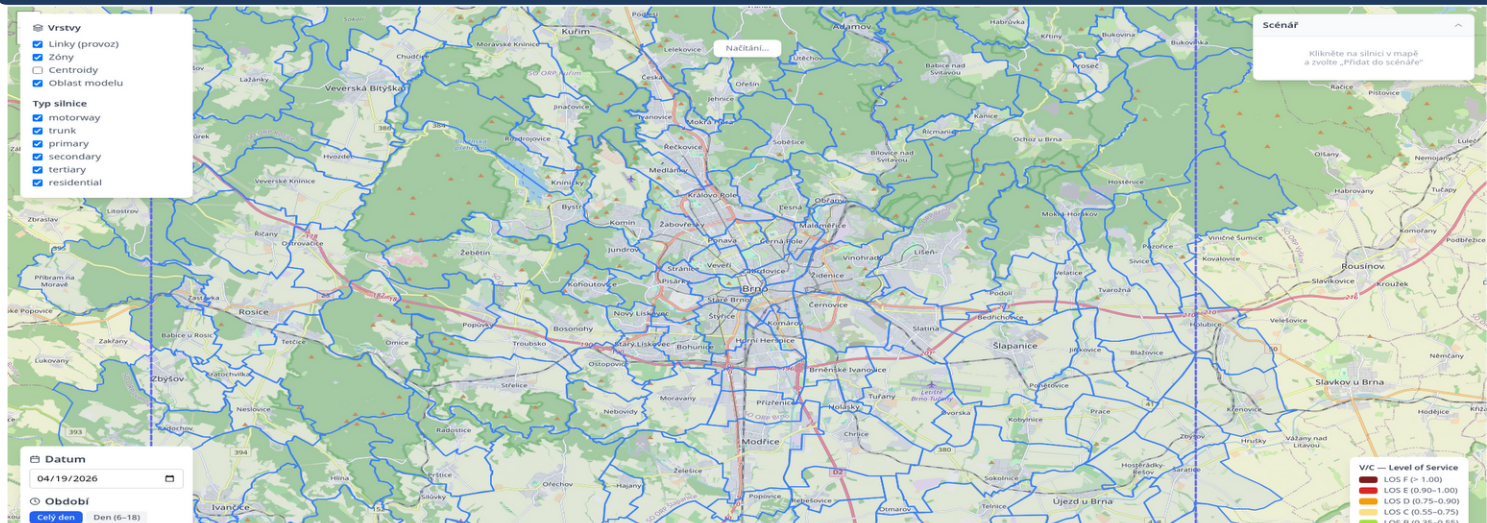
Shared inputs → in-memory graph mod → per-link delta

Figure 5: Baseline Traffic (V/C Ratio)



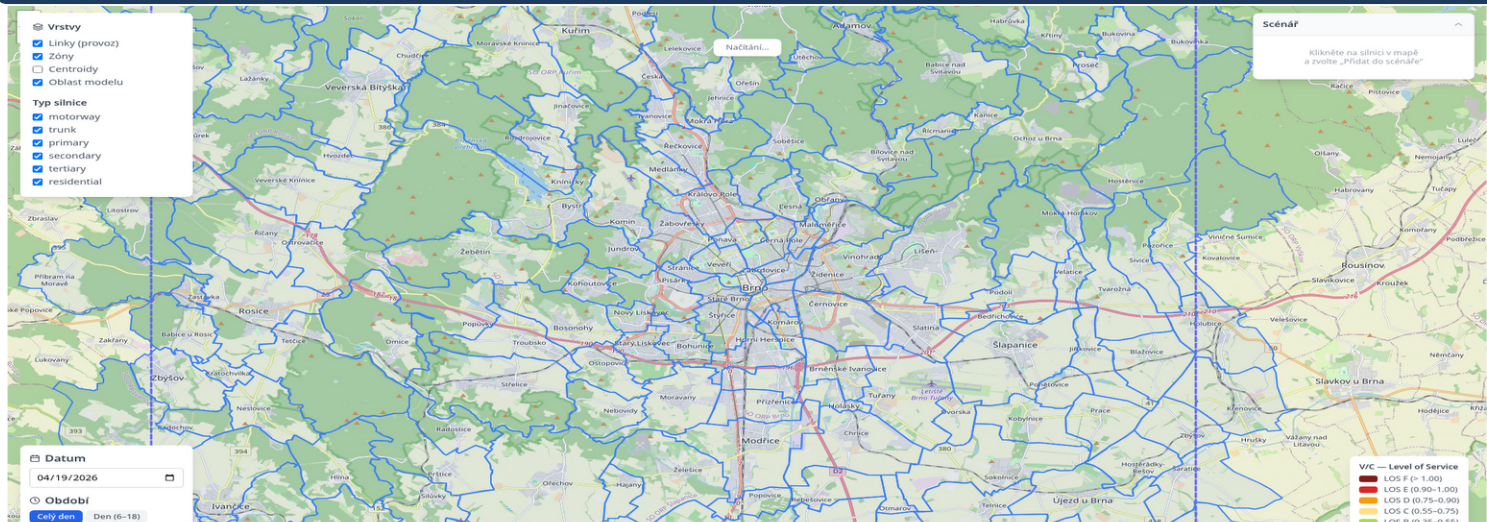
LOS colored links: green (A) → dark red (F)

Figure 6: Scenario — Closure Applied



Interactive link selection, full/fane closure types

Figure 7: Delta — Traffic Redistribution



Red = increase, Blue = decrease, width ∝ |Δvol|

Figure 8: Calibration & Reports

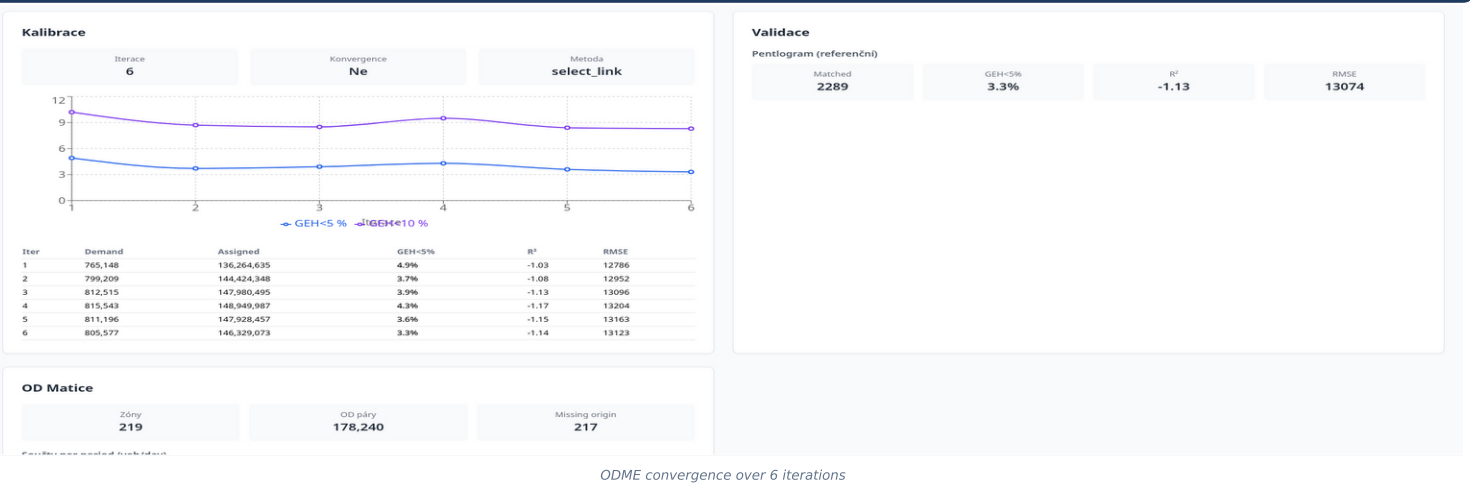
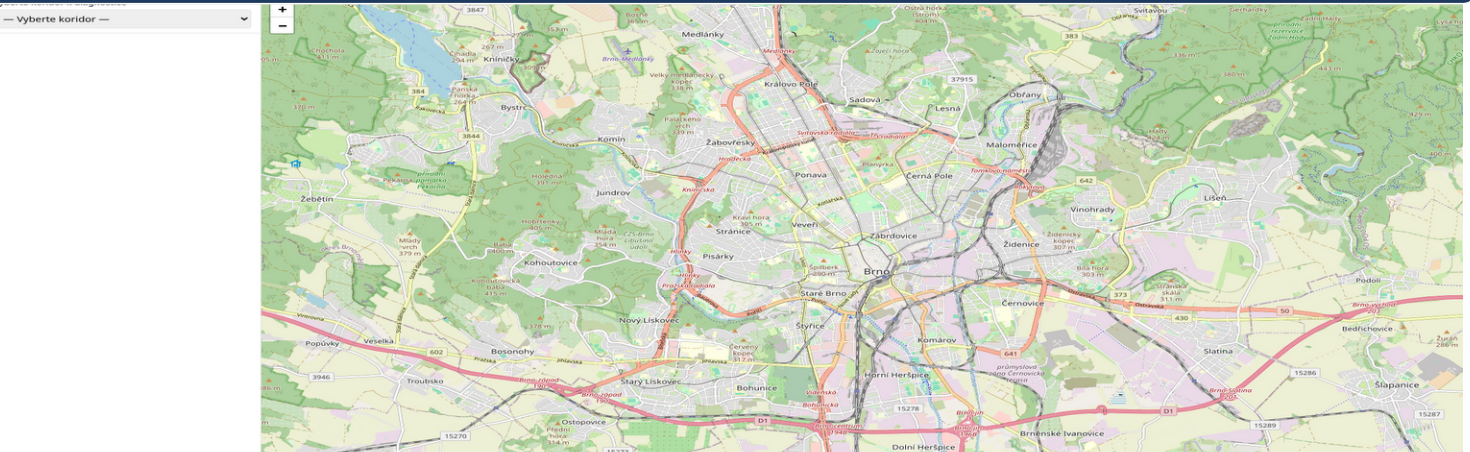


Table 1: Model Metrics

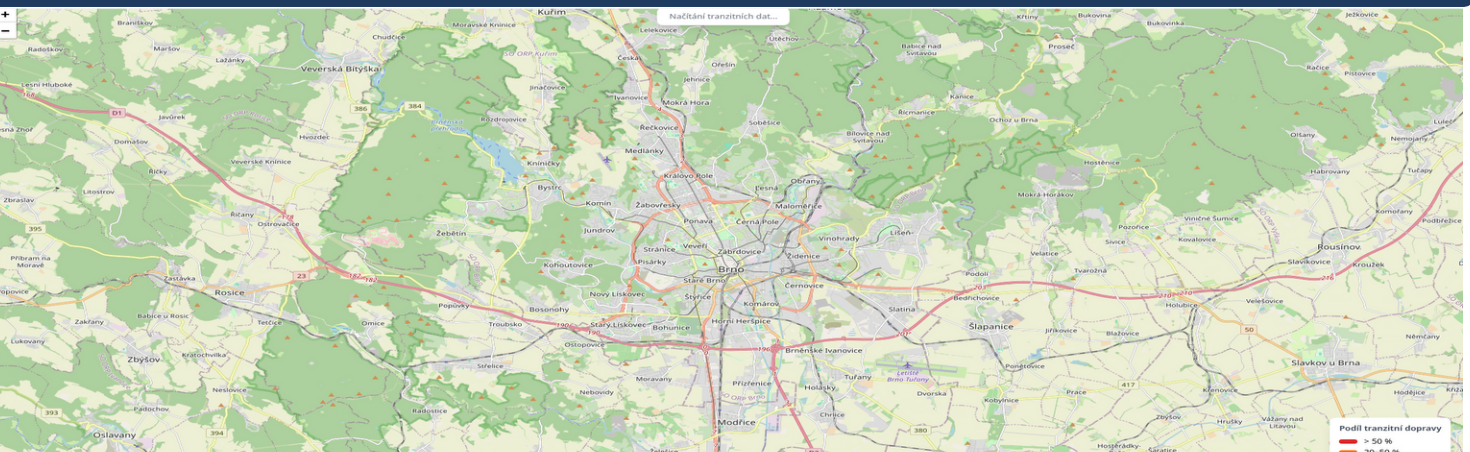
Metric	Value
Network	52k links, 47k nodes, 219 zones
Daily demand	765,148 vehicles
Commuting	691,685
Through-traffic	73,463
Count stations	2289 matched
Overall bias	-2.5%
D1 east ratio	1.08
Krenova ratio	1.19
ODME iterations	6

Figure 9: Corridor Diagnosis



Free-flow vs forced path, time/distance comparison

Figure 10: Through-Traffic Analysis



External-through share, gateway markers, screenlines

KEY CONTRIBUTIONS

• Reproducible 16-step open-source pipeline (OSM + Czech open datasets only)

• National supernetwork for data-driven through-traffic (6117 places, detour-filtered)

• Spies-style ODME calibration with select-link screenlines + gateway damping

• Non-destructive scenario engine for interactive What-if closure analysis

• Rich diagnostic API: corridor explanation, bias clustering, through-traffic overlay